

Relation Algebra

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ \neg \times \text{Digit}(\text{val}^\circ) \text{snoc}$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the

rest

(a)

(b)

(c)

(d)

(e)

(f)

(g)

(a) $\text{val} = ([\text{embed}, \text{op}])$ — definition

(b) $(\varphi) = \alpha^\circ F(\varphi) \varphi$ — catamorphisms

(c) $(RS)^\circ = S^\circ R^\circ$ — converse

(d) $F(R)^\circ = F(R^\circ)$ — $F = (\neg \times \text{Digit})(\text{Digit}^{++})$, definition of F

(e) $\alpha = [\text{wrap}, \text{snoc}]$

(f) $(1 + S)[P, Q] = [P, SQ]$ at $S = (\neg \times \text{Digit})(R)$ — coproduct

(g) $[g, h]^\circ [P, Q] = g^\circ \text{Puh}^\circ Q$ — coproduct

§6.2. $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

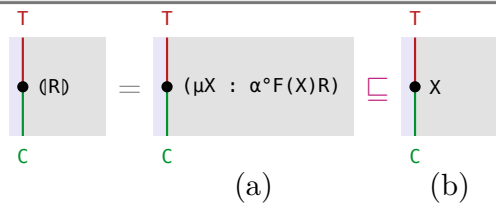
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

$$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow (R) \sqsubseteq X$$

(6.2c)

a relation X that contains one unfolding of the fold's recursion at X contains the fold



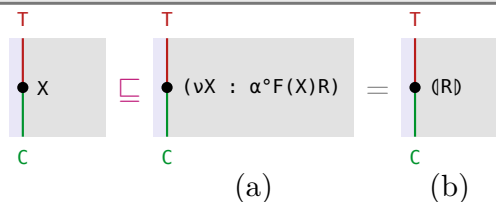
(a) (R) the least fixed point — Theorem 6.1, (6.2b)

(b) $\alpha^\circ F(X) R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq (R)$$

(6.2d)

a relation X contained in one unfolding of the fold's recursion at X is contained in the fold



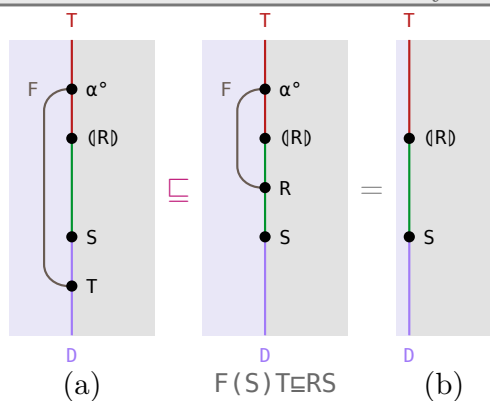
(a) $X \sqsubseteq \alpha^\circ F(X) R$, ν above every postfix point

(b) (R) the greatest fixed point

$$F(S) T \sqsubseteq R \Rightarrow (T) \sqsubseteq (R) S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



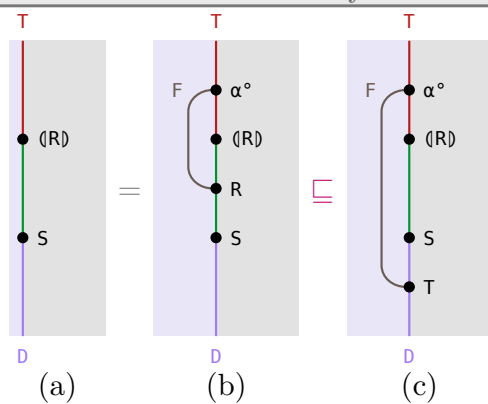
(a) (6.2) at $X \triangleq (R) S$

$F(RS) = F(R) F(S)$ — $\triangleq$ relator

(b) $F((R)) R = \alpha(R)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce

$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2f)

 if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T

 (a) (6.3) at $X \triangleq (R)S$

 (b) $\alpha(R) = F((R))R$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce

 (c) $RS \sqsubseteq F(S)T$
 $F(R)F(S) = F(RS)$ — $\triangleq$ relator

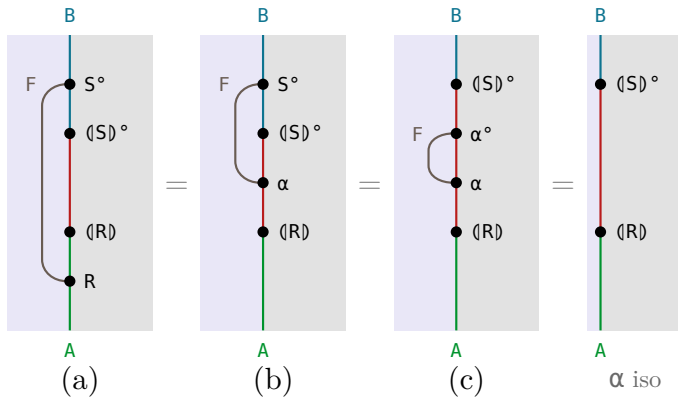
§6.3. $(S)^\circ (R) = (\mu X : S^\circ F(X)R)$

$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

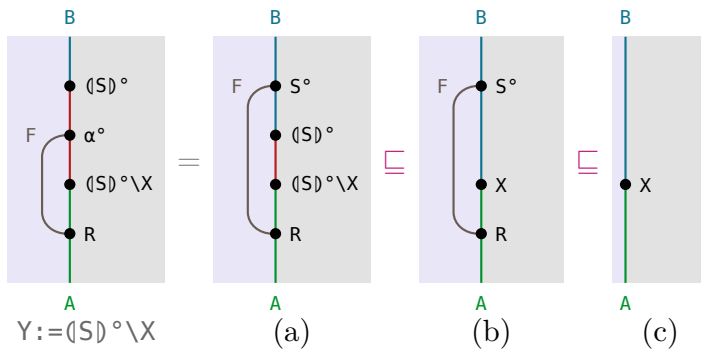
$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

 hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term $\mathbb{S}^\circ$ represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial

 (a) the body at $(\mathbb{S})^\circ(\mathbb{R})$
 $F(RS) = F(R)F(S) \triangleq \text{relator}$

 (b) $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq \text{reduce}$

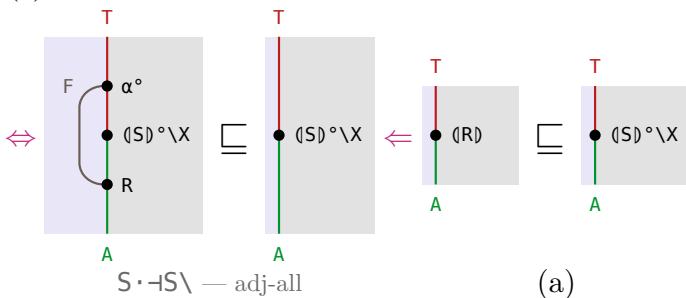
 (c) $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$

$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

 hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

 $Y := (\mathbb{S})^\circ \setminus X$

 (a) $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ)$
 $F(RS) = F(R)F(S) \triangleq \text{relator}$

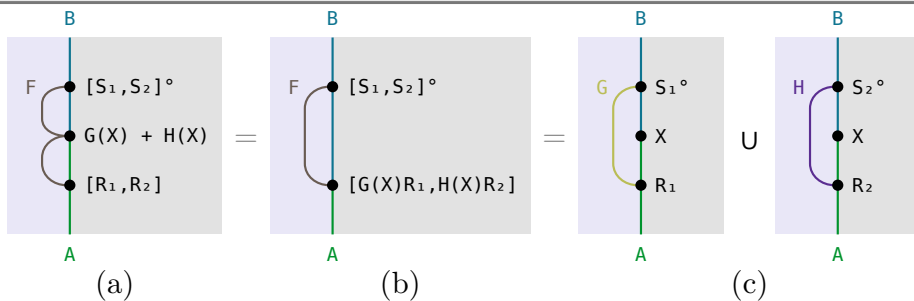
 (b) $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$

 (c) $\mathbb{S}^\circ F(X)R \sqsubseteq X$

 (a) $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.2b), adj-all}$

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \implies ([S_1, S_2])^\circ([R_1, R_2]) = (\mu X : S_1^\circ G(X)R_1 \cup S_2^\circ H(X)R_2)$$

(6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results


 (a) the body of Theorem 6.2 at $F(X) = G(X) + H(X)$

 (b) $(P+Q)[R_1, R_2] = [PR_1, QR_2] \triangleq \text{coproduct}$

 (c) $[S_1, S_2]^\circ[P, Q] = S_1^\circ P \cup S_2^\circ Q \triangleq \text{coproduct}$

§6.4. Fast exponentiation and modulus computation

§6.5. Unique fixed points

§6.6. Sorting by selection

§6.7. Closure